

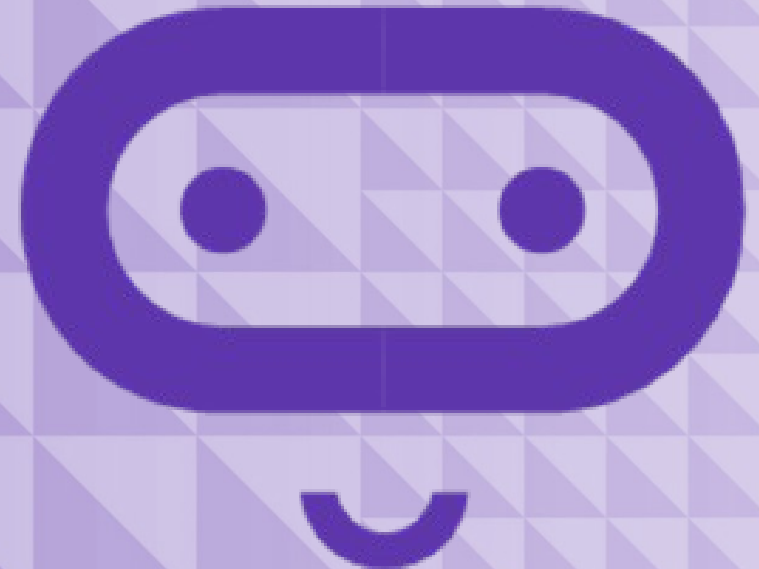
micro:bit

Introduction to Computer Science

Unit 4: Conditionals

Lesson A

Conditionals in daily life



What we'll learn in Lesson A

- Conditional statements
- How conditionals are used to tell a computer when to do something

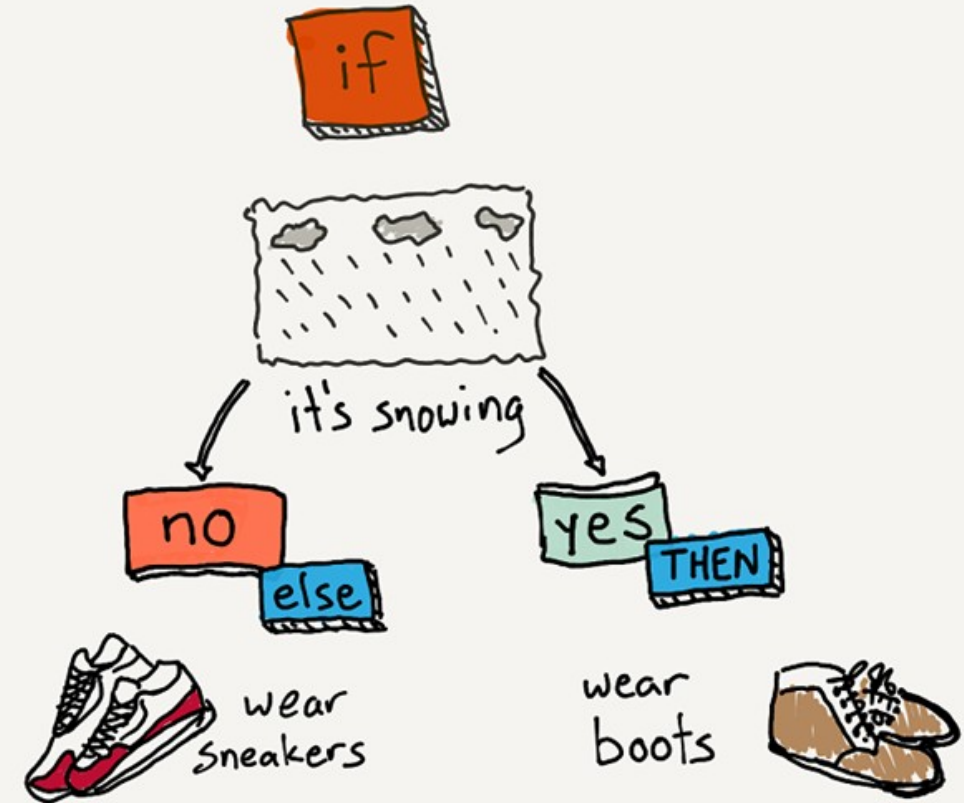
Conditionals

What is a conditional?

- A rule (or a condition) that must be met for an action to be carried out
- Written as an **IF... THEN...** or **IF... THEN... ELSE...**

Conditional statements

- “**IF** you clean your room,
THEN you can go out with your friends.”
- “**IF** your homework is done,
THEN you can play video games.”
- “**IF** you do your chores all week,
THEN you get your allowance,
ELSE you are grounded.”



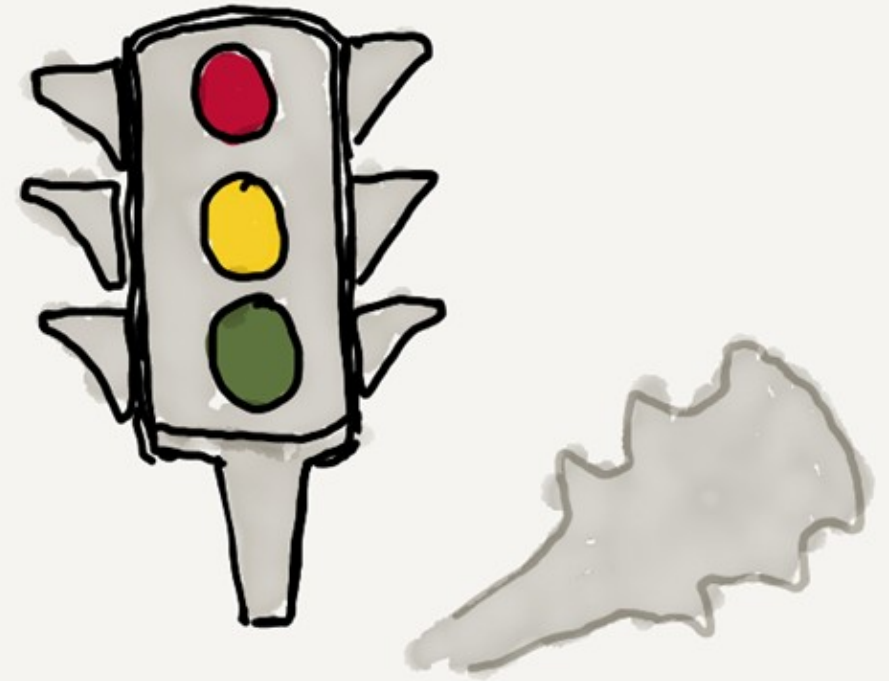
Red light, green light

Directions:

- Everyone lines up on one side of the room.
- The object of the game is to get across the room.
- But in order to do this, you need to listen to the conditions—and only perform the action if the condition is true for you.

Examples:

- “**IF** you are wearing something green, **THEN** take a step forward”
- “**IF** your birthday is this month, **THEN** then take a step forward, **ELSE** take a step backward”



Debrief discussion

- Are there some conditions that can be evaluated as something other than True or False (maybe, sometimes)?
- How might Conditional Statements be used to give instructions to a computer?

Examples:

- “**IF** someone presses the power button, **THEN** turn on”
- “**IF** the login box is blank, **THEN** tell the user they need to type in their name, **ELSE** log in”



What we learned in Lesson A

- Conditional statements
- How conditionals are used to tell a computer when to do something



Next time: Lesson B

- Code the micro:bit to play Rock, paper, scissors
- Assess our learning with a quiz



Lesson B

Rock, paper, scissors

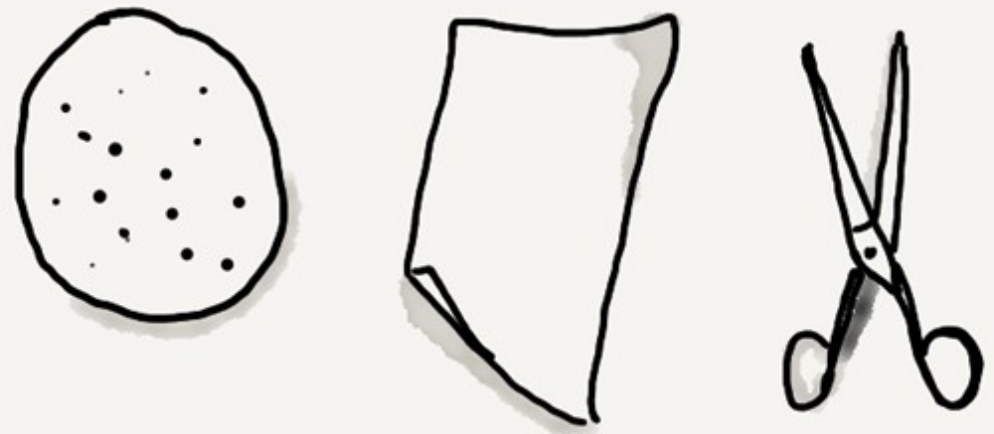


What we'll learn in Lesson B

- Code a game with conditionals
- Review how much we've learned

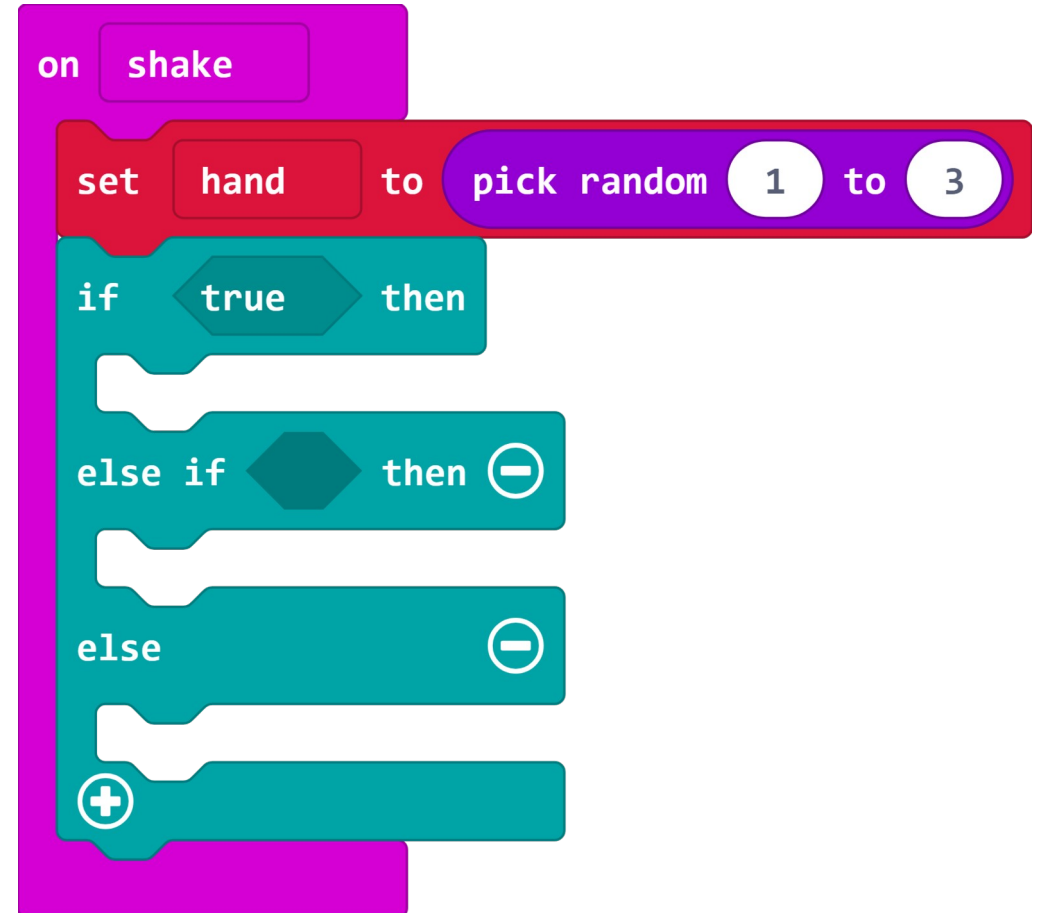
Unplugged activity: Rock, paper, scissors

1. What are the rules of the game?
2. What are the conditionals?
3. Work in pairs to code a “Rock, paper, scissors” game.
4. Start with pseudocode (instructions in English)
5. Mod the game to add more options, like rock, paper, scissors, lizard, Spock. Get creative!



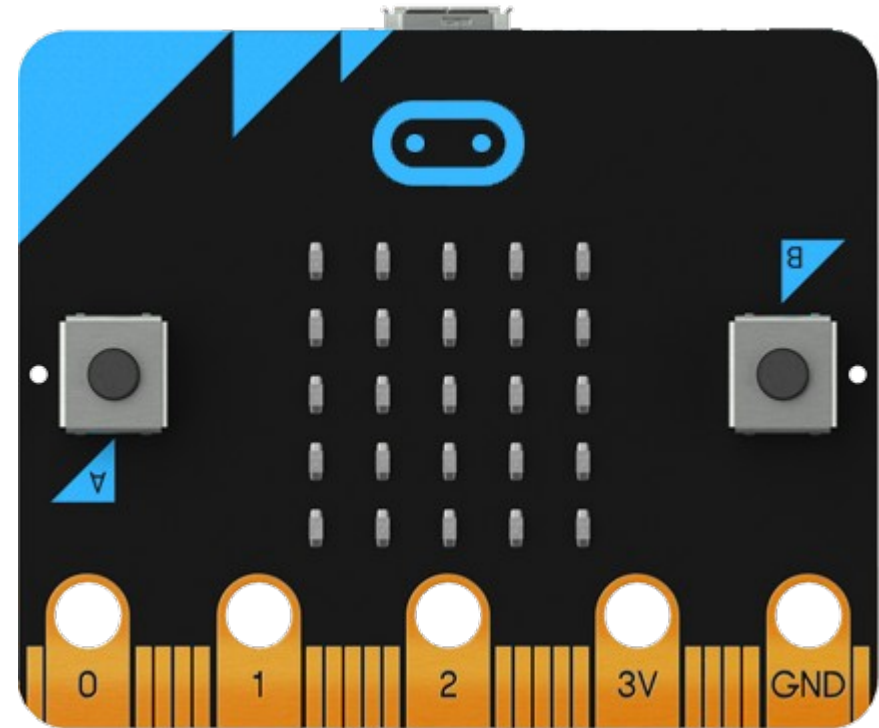
Hints

- Use **On Shake** to start the game
- Use **Pick Random** to choose rock, paper or scissors
- Use an **If... Then... Else if... Else...** block for three choices

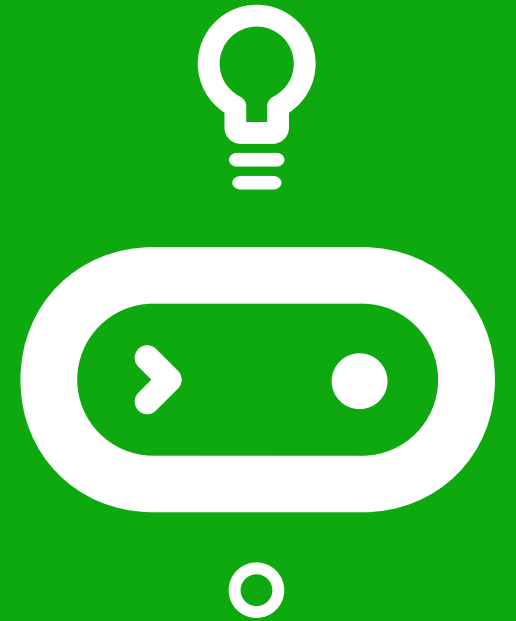


Modifications (Mods)

- Add a way to keep score.
- Add more options and more game rules in addition to rock, paper, scissors (e.g., lizard, Spock)
- Use other micro:bit Inputs (buttons, pins)



Time for a quiz!



What we learned in Lesson B

- Coded a game with conditionals
- How much we've already learned!



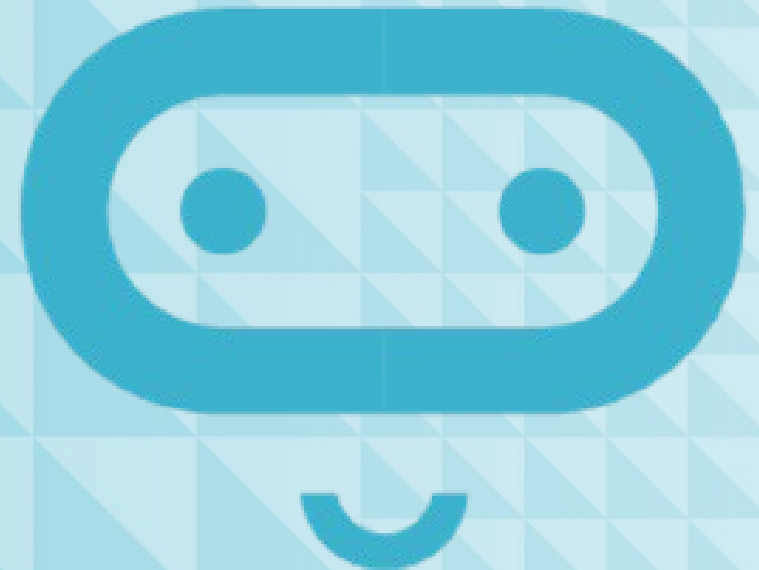
Next time: Lesson C

- Create an original board game



Lesson C

Create a board game



What we'll learn in Lesson C

- Create an original board game using conditionals and your micro:bit

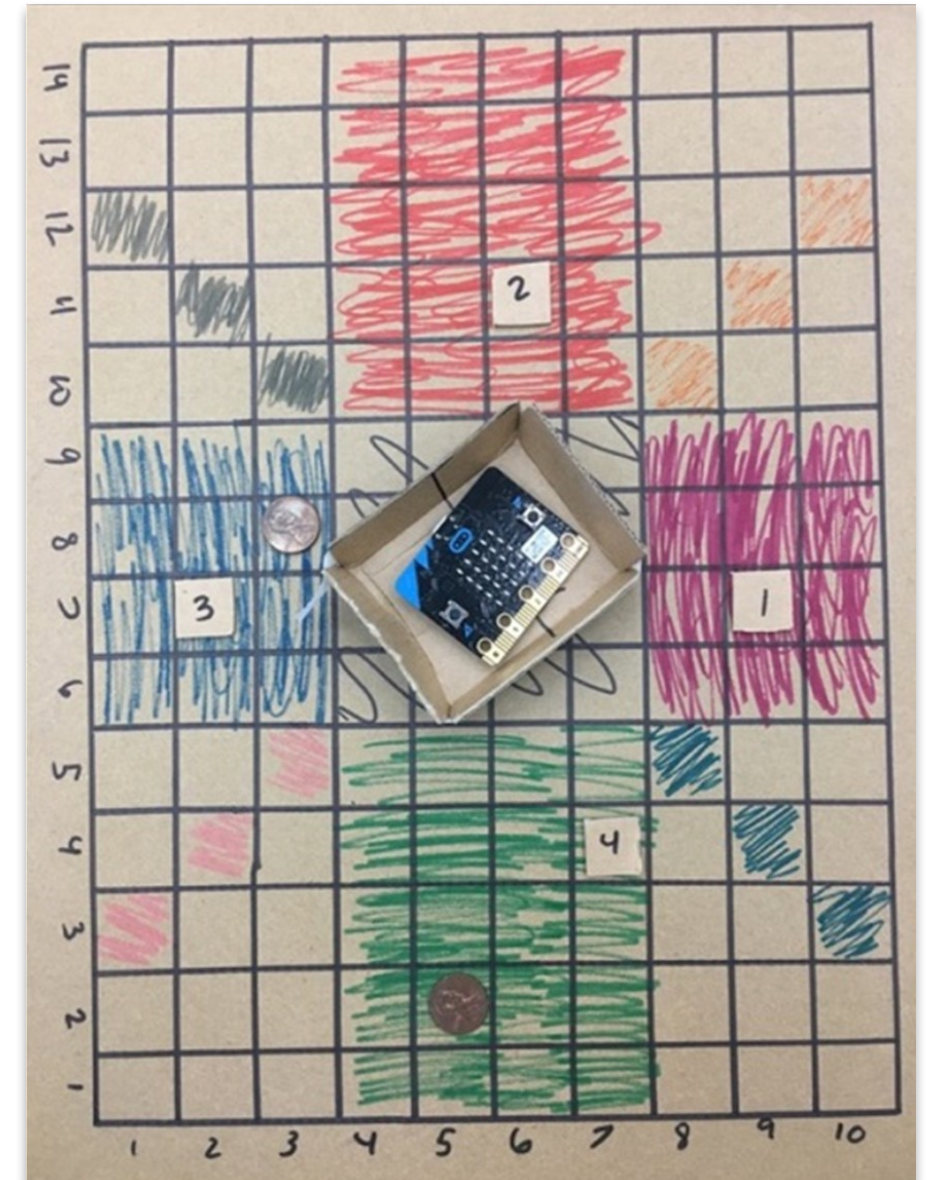
Invent a game

- Create an original board game that incorporates conditionals and your micro:bit to play.
- Work together to come up with:
 - A set of written rules (how to play)
 - A game board
 - A program for the micro:bit that works in conjunction with the game board and/or game pieces



Some project ideas

- **Game Rules** – Write these as a set of Conditional Statements:
 - **IF** Heads, **THEN** move 3 spaces, **ELSE** lose a turn.
- The micro:bit might:
 - Simulate the results of a battle between two game pieces.
 - Randomly point a player in a different direction of travel.
 - Generate a result based on its current incline.
 - Display a dynamic score.
- Let your imaginations run wild!



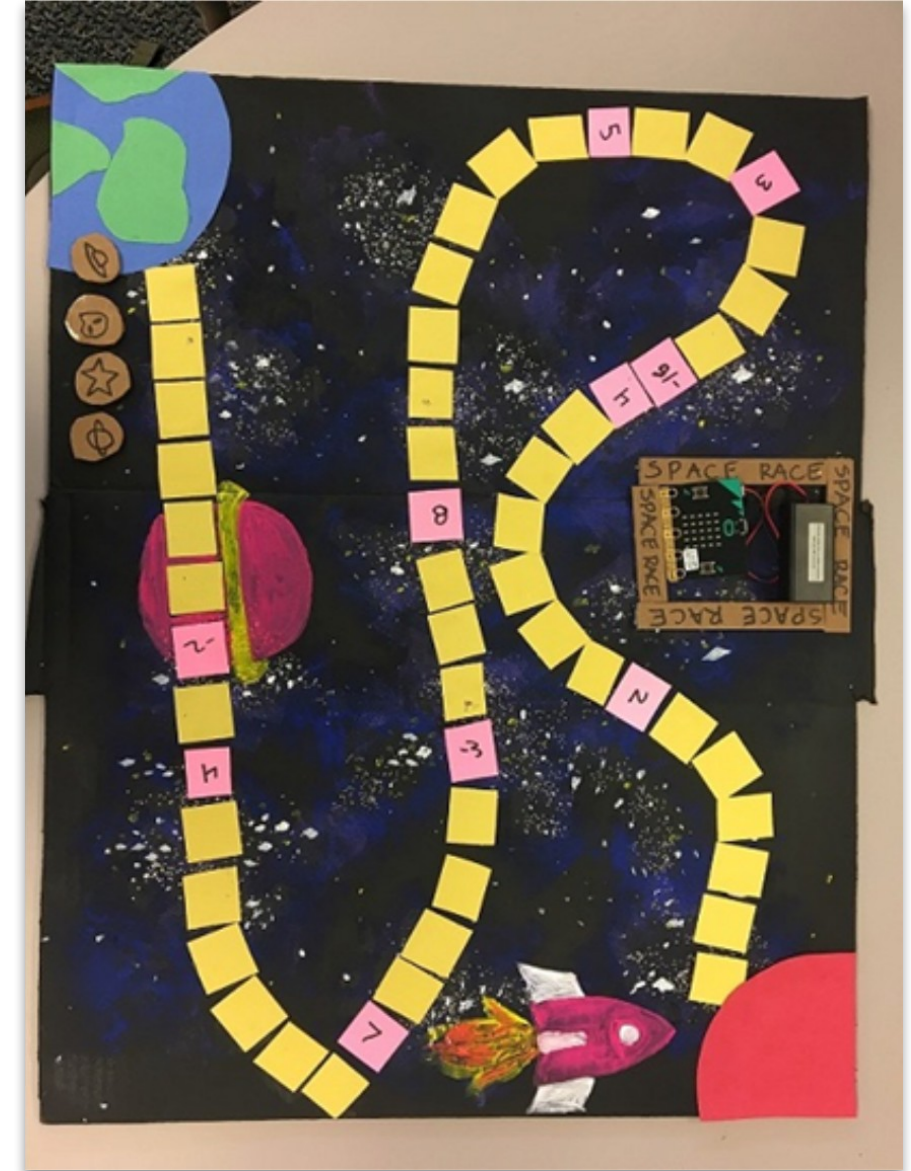
Board game example

Space Race by K. and S.

- **How to win:** Starting from Earth, your goal is to progress to Mars. The first person to reach Mars is the winner.

Rules:

- Shake the micro:bit to randomize how far you get to advance.
- If you land on a pink square, press “B” on the micro:bit until your previous roll number appears. Then press A and B at the same time to see whether or not you move based upon the number on the square.
- Up to four players.



Project expectations

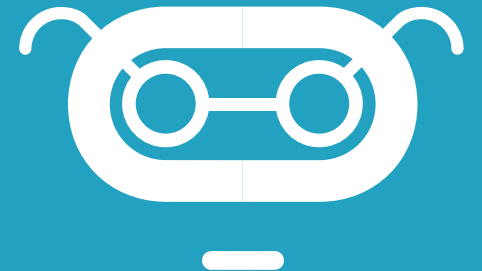
Make sure your project meets these specifications:

- All **games rules** should be clear and complete.
- The **game board** should be complete, neat, and fit the theme of the game.
- The **micro:bit** must be an integral part of the game and the board.
- The program correctly uses **conditionals**, compiles and runs as intended, and includes comments.
- **Photo documentation** that includes photos of the game board, code, and captions.
- Provide the written **Reflection Diary** entry that includes your brainstorming ideas, construction, programming, and beta testing (which we'll talk about after you complete your project).

Reflection diary

Explain how you decided, as a pair, on your particular board game idea.

- What was something that surprised you about the process of creating this game?
- Describe a difficult point in the process of designing this game and explain how you resolved it.
- What feedback did your beta testers give you? How did that help you improve your game? What were the Conditionals that you used as part of your game rules?
- Publish your MakeCode program and include the link.



What we learned in Lesson C

- Created an original board game using conditionals and your micro:bit

